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Alleviation of volatile fatty acids inhibition in anaerobic digestion of swine manure with nano-bubble water supplementation

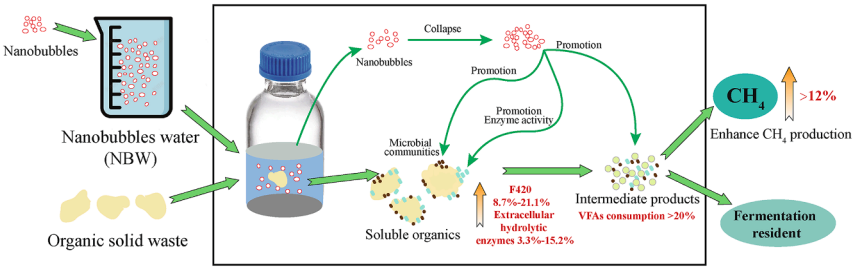
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HIGHLIGHTS

- Volatile fatty acids (VFAs) inhibition was alleviated by nano-bubble water (NBW).
- NBW supplementation promoted buffering capacity under VFAs inhibition.
- CH_4 production was enhanced by NBW supplementation under VFAs inhibition.
- Activity of hydrolases and coenzyme F420 were increased with NBW addition.
- Enriched microbial community and reinforced metabolic pathway were observed.

GRAPHICAL ABSTRACT



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ABSTRACT

Nano-bubble water (NBW) was applied to anaerobic digestion (AD) to alleviate volatile fatty acids (VFAs) inhibition, improve the buffering capacity and CH_4 production in this work. Results indicated that NBW accelerated the consumption of VFAs and prevented inhibition due to VFAs accumulation. Additionally, NBW facilitated a rapid increase in partial alkalinity (PA) and total alkalinity (TA) as well as a corresponding rapid decrease in intermediate alkalinity (IA)/PA and VFA/TA, thereby improving buffering capacity and alleviating VFAs inhibition. Moreover, CH_4 production improved by more than 12.2% by NBW. Similarly, the activities of the extracellular hydrolases and coenzyme F420 increased. Besides, NBW increased the abundance of microbial community and strengthened the metabolic pathways of hydrogenotrophic methanogens, which could be the intrinsic mechanism by which NBW alleviated VFAs inhibition, improved system stability, and increased CH_4 production. This study demonstrates that NBW supplementation can be an effective method for mitigating frequent VFAs inhibition.

1. Introduction

Large amounts of agricultural waste such as livestock, poultry manure, and crop straw are generated annually worldwide. China alone produces more than 4.3 billion tons of agricultural waste each year (Aravani et al., 2022). If agricultural wastes are not effectively treated,

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they not only pollute the environment but also deplete biomass resources. Agricultural waste poses both environmental pollution challenges and opportunities to convert waste into biomass energy. Anaerobic digestion (AD), a time-tested waste treatment method, is an effective means of addressing the dual challenges of environmental pollution and resource scarcity (He et al., 2022; Liu et al., 2023). Globally, carbon emissions and related global warming have become a threat to the world, mainly due to the widespread use of fossil fuels and the uncontrolled generation of solid waste. AD is a sustainable technology that has been recognized as a low carbon technology that complements waste management and renewable energy. AD provides many advantages, including clean energy production, odour control, and pathogen inactivation (Zhang et al., 2022; Cui et al., 2022), making it a dual-purpose solution for environmental protection and resource recycling when applied to agricultural waste treatment.

Volatile fatty acids (VFAs) is a type of fatty acid, generally composed of organic acids with carbon chains ranging from 1 to 6 atoms, including acetic acid, propionic acid, etc. High concentrations of VFAs can be toxic to microorganisms during the AD process, inhibiting acetate consumption during the methanogenesis phase and resulting in reduced CH_4 production efficiency (Zhao et al., 2018). The optimal pH range for methanogenesis is typically between 6 and 8. When the pH drops below 6, the activity of methanogens decreases due to the highly acidic environment. During VFAs inhibition, the lowered pH further suppresses methanogenic activity. The inhibition of VFAs reduces the efficiency of agricultural waste treatment, impedes clean energy production, and severely limits its application in the field of agricultural waste resource utilization (Zhao et al., 2022). For example, in the AD process of kitchen waste, VFAs can easily accumulate, and high concentrations of VFAs inhibit the AD process (Yan et al., 2023a). Therefore, VFAs inhibition impedes the efficient treatment of agricultural waste, necessitating the adoption of scientifically proven methods to eliminate inhibition and improve the efficiency of agricultural waste treatment. There are many methods available to alleviate VFAs inhibition during AD process, such as the addition of alkaline agents (Zhou et al., 2023), trace elements (Lim et al., 2022), biochar (Zhu et al., 2022), and the combined use of microbial enhancers and activated carbon (Zhao et al., 2022). However, these traditional methods have certain limitations, such as secondary pollution from chemical agents, high energy consumption, high costs, and complexity of operation, which have not been widely applied in the field of agricultural waste resource utilization (Lim et al., 2022). Therefore, there is an urgent need to discover new methods that can effectively eliminate VFAs inhibition in AD process, without causing secondary pollution, simple operation, and cost-effective.

Whether it is wet, semi-dry, or dry AD process, water is required for the processes of mass transfer, energy conversion, and microbial electron transfer. If the properties of fermentation liquid can be altered to improve mass transfer efficiency and microbial activity, VFAs inhibition may be alleviated. Previous studies on water in the AD process had primarily focused on water content, specifically investigating the influence of total solids (TS) (Kizito et al., 2022; Abid et al., 2021). Although TS had an effect on the AD process, dilution with water did not fundamentally address the issue of VFAs inhibition and wasted water resources. Many previous studies in this field had focused on the effect of water content on the AD process. However, few studies had examined the impact of different water types and characteristics on the AD process. It remains unclear whether VFAs inhibition can be mitigated by altering the types and characteristics of water. Nano-bubble water (NBW) refers to water containing a high concentration of nanometer-sized bubbles, ranging from a few nanometers to several hundred nanometers in diameter (Hou et al., 2022). This technology debuted in the late 1990 s and gained momentum in Japan in the early 21st century. The main characteristics of NBW include the reduced friction, the negative or positive zeta potentials, increased gas solubility in liquids, and the generation of free radicals (Liu et al., 2013). Liu et al., (2019) found that NBW had superior adsorption compared to deionized water (DW). Additionally, the collapse of nanobubbles would produce a significant amount of reactive oxygen species (ROS), which can enhance the degradability of various recalcitrant organics (Khan et al., 2014). Furthermore, the supplementation of NBW could improve mass transfer efficiency (Temesgen et al., 2017). Moreover, NBW production is simple to operate with low energy comsuption, and emits no chemical pollutants. Despite its widespread application in various fields, such as agriculture (Sun et al., 2022), environmental protection (Jia et al., 2023), and medicine (Aoki et al., 2022), the potential of NBW has not been fully explored. Recently, NBW has been found to be able to promote CH_4 production during the AD process (Yang et al., 2019; Fan et al., 2020).

Considering the unique characteristics of NBW and the fact that it can play a positive role in the AD process, it would be of great promise in alleviating VFAs inhibition. Therefore, the goal was to study the alleviating effects of NBW on different degrees of VFAs inhibition and to uncover the underlying mechanisms. This study aimed to reduce VFAs inhibition and accelerate the AD process by developing a new pollution-free, low-energy and simple method. This study is the first attempt to use NBW to alleviate VFAs inhibition in AD. The results of this study suggest a new approach for alleviating VFAs inhibition and provide an innovative solution to the common inhibition phenomenon.

Table 1
Characteristics of raw swine manure and inoculum used in this study.

Parameters	Unit	Inoculum Raw swine manure
Total solids	%	1.35 ± 0.08 27.20 ± 0.79
Volatile solids	%	0.87 ± 0.04 22.82 ± 0.58
Total ammonia nitrogen	mg/g	0.43 ± 0.04 4.02 ± 0.27
Soluble total carbon	mg/g	ND 36.12 ± 1.08
Soluble inorganic carbon	mg/g	ND 3.96 ± 0.13
Soluble total organic carbon	mg/g	ND 32.15 ± 0.70
Volatile fatty acids	mg/g	ND 31.48 ± 0.91
Total alkalinity (as $CaCO_3$)	mg/g	1.81 ± 0.10 15.09 ± 0.64
Soluble proteins	mg/g	0.29 ± 0.02 12.11 ± 0.29
Soluble carbohydrates	mg/g	0.09 ± 0.01 5.90 ± 0.36
C	%/TS	39.09 45.31
H	%/TS	5.17 6.04
N	%/TS	4.14 4.25
S	%/TS	1.09 1.66
C/N ratio	–	9.44 10.66
pH	–	7.46 ± 0.04 6.45 ± 0.05

ND (Not Determined).
Data in the table are expressed as Means ± SD.

2. Materials and methods

2.1. NBW

An NBW generator (Yuchuang Environmental Technology Co., Ltd., Shandong, China) was used to produce NBW. Air-NBW was used in this experiment because it was less expensive and more practical than O_2-NBW, N_2-NBW, H_2-NBW, etc. Air (natural air, oxygen content is about 21 %) was introduced into the NBW generator, and the specific preparation process followed the methods described in the relevant studies (Tang et al., 2022; Chen et al., 2022). Following completion of the preparation process, the obtained NBW was used for subsequent experiments.

2.2. AD experiments

2.2.1. Raw materials

In this study, original swine manure samples were collected from a pig-farming facility in Guigang City, Guangdxi Province, China, and used as substrates. These samples were thoroughly sorted to remove impurities (such as feed residues, plastics) before being evenly mixed and stored. Digested sludge was collected from Guangzhou City, Guangdong Province, China. To acclimate the inoculum, the digested

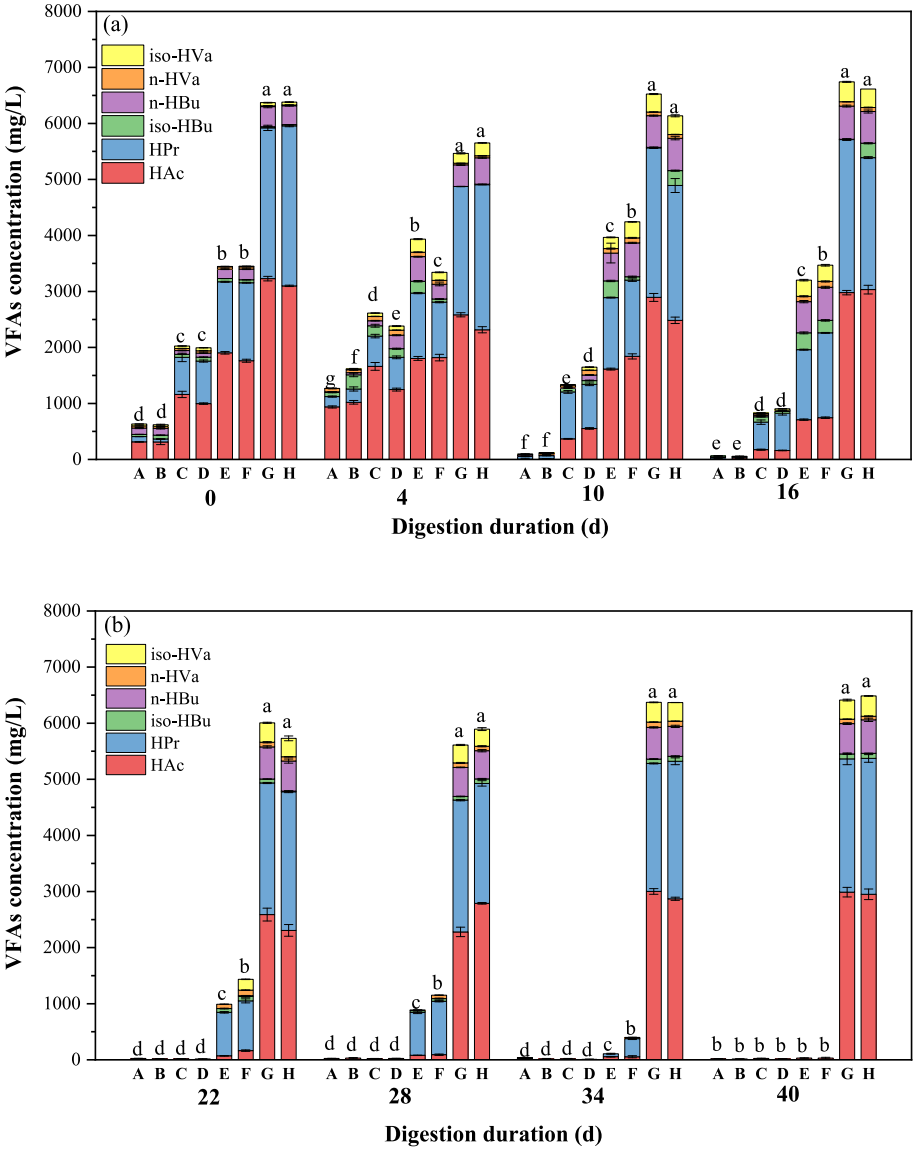


Fig. 1. VFAs concentration variation in AD process. (a) VFAs concentration in each group on day 0, day 4, day 10, and day 16. (b) VFAs concentration in each group on day 22, day 28, day 34, and day 40. The alphabets under X-axis areas: A-R(0)$_{NBW}$, B-R(0)$_{Control}$, C-R(0.125)$_{NBW}$, D-R(0.125)$_{Control}$, E-R(0.25)$_{NBW}$, F-R(0.25)$_{Control}$, G-R(0.5)$_{NBW}$, H-R(0.5)$_{Control}$. Within these groups, 0.5 mL, 0.25 mL, 0.125 mL, and 0 mL of VFAs (1000 mg/L) were added in R(0.5)$_{Control}$/R(0.5)$_{NBW}$; R(0.25)$_{Control}$/R(0.25)$_{NBW}$; R(0.125)$_{Control}$/R(0.125)$_{NBW}$; R(0)$_{Control}$/R(0)$_{NBW}$, respectively. NBW stands for nano-bubble water. One-way analysis of variance (ANOVA) followed by Tukey's post-hoc test was applied for the statistical analysis of VFAs. The different letters (a, b, c, …) indicate significant difference at the 5 % level ($p < 0.05$).

sludge was cultivated using 1 L bottles with the raw swine manure samples (collected from the same pig-farming facility in Guigang City, Guangdxi Province, China) about one month until no biogas production. The swine manure collected was added into the 1 L bottles to achieve a feed-to-microorganisms (F/M) ratio of 1:1 (volatile solid (VS) basis) at the beginning of this inoculum domestication process. The initial pH in reactor was adjusted to 7.00 ± 0.05 using 2.0 M NaOH or HCl. The 1 L bottles used in this domestication process were placed into a temperature- controlled incubator (35 ± 2 °C). After the acclimation process, the inoculum obtained was used in subsequent experiments. The characteristics of the inoculum and swine manure are presented in Table 1.

2.2.2. AD experiments

In this study, a series of identical 250 mL Schott Duran serum bottles were used as the reactor in this AD experiments, each with a working volume of 200 mL. At the beginning of the AD experiment, all 250 mL reactors were loaded with substrate (collected from the same pig-farming facility) and inoculum. The total loading of substrate and inoculum was 100 g. The seeding ratio (feed-to-microorganisms (F/M) ratio) in each reactor was maintained at approximately 1:1 (VS basis) and the swine manure collected from the same pig-farming facility was used as the substrate in this work. Initially, varying volumes of the VFAs standard solution (1000 mg/L) were added to 250 mL bottles to simulate VFAs inhibition. This study divided its participants into eight groups: R(0.5)$_{Control}$/R(0.5)$_{NBW}$; R(0.25)$_{Control}$/R(0.25)$_{NBW}$; R(0.125)$_{Control}$/R(0.125)$_{NBW}$; R(0)$_{Control}$/R(0)$_{NBW}$, where 0.5 mL, 0.25 mL, 0.125 mL, and 0 mL of VFAs were added, respectively. To avoid introducing other chemical agents that could alter the microbial structure and to minimise interference from other factors, the pH was not adjusted throughout the experiment. 100 mL DW or NBW was added separately to the $R_{Control}$ and R_{NBW}, respectively. Additionally, to ensure the initial anaerobic environment, nitrogen gas was flushed into all groups for 3 min before being placed in incubator (35 ± 2 °C). The reaction was stopped after 6 days of no gas production.

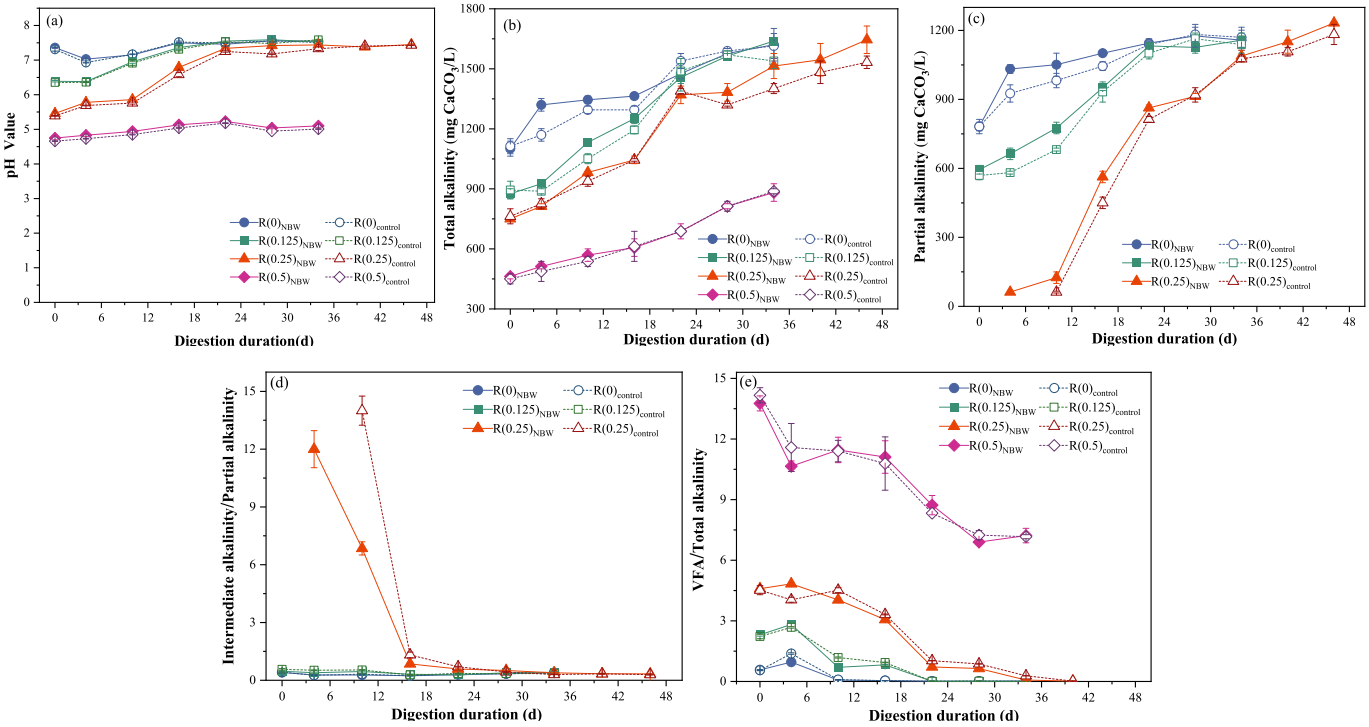


Fig. 2. Changes in Stability Parameters. (a) pH value. (b) Total alkalinity. (c) Partial alkalinity. (d) Intermediate alkalinity/Partial alkalinity. (e) VFA/Total alkalinity.

2.3. Analysis methods

In this experiment, biogas production was measured every two days using a 60 mL airtight syringe. CH_4 and other gas components were measured by gas chromatography (GC-8A/TCD, SHIMADZU Co., Japan). The VFAs content was determined by gas chromatography (GC-8A, SHIMADZU Co., Japan). The standard methods (APHA 2012) were used in this study to measure TS, VS, and total ammonia nitrogen (TAN). Soluble carbohydrates and proteins were determined according to the previous research (Fan et al., 2020). Coenzyme F420 was determined using previous studies (Hou et al.,2022; Delafontaine et al., 1979). Previous literature reports were used to determine intermediate alkalinity (IA), total alkalinity (TA), and partial alkalinity (PA) (Sun et al., 2016; Banks et al., 2012). Specifically, the basicity of the sample was determined by acid-base titration, in which PA and TA were measured at pH 5.75 and 4.3, respectively. IA was measured by pH titration from 5.75 to 4.3.

On day 4, protease, alkaline phosphatase (ALP), α-glucosidase, and acid phosphatase (ACP) were measured based on previous studies (Yuan et al., 2017). Genomic DNA was extracted from the sludge samples at the end of digestion using the E.Z.N.A. Soil DNA Kit (OMEGA). The concentration and purity of extracted DNA samples were determined using a NanoDrop 2000 spectrophotometre (NanoDrop Technologies). 16S rRNA were amplified using different sets of primers for the analysis of bacterial communities (338F('5-ACTCCTACGGGAGGCAGCAG-3′)/806R (5′-GGACTACHVGGGTWTCTAAT-3′)) and archaeal communities (524F10extF (5′-TGYCAGCCGCCGCGGTAA-3′)/Arch958RmodR (5′YCCGGCGTTGAVTCCAATT-3′)) by using BMK Cloud Platform (Tang et al., 2024).

2.4. Statistical analysis

All experiments were conducted in triplicates and data are shown as the mean or mean ± standard deviation. An analysis of variance (ANOVA) followed by Tukey's post-hoc test was conducted to identify the significance of the results.

3. Results

3.1. Changes in VFAs concentration

The conversion of macromolecules into soluble substrates is the rate limiting step. VFAs are important intermediate products of the AD process. In this study, the concentrations of VFAs was measured as shown in Fig. 1. The VFAs concentration in R(0) increased on the 4th day. The increase in the VFAs concentration on the 4th day was likely due to the consumption of soluble substrates during the hydrolysis stage, leading to the production of VFAs by microorganisms. Additionally, on the 4th day, the average concentration of VFAs in $R(0)_{NBW}$ was 1269 mg/L, representing decrease of 27.7 % compared to its corresponding control group. This was attributed to NBW, which accelerated the consumption of VFAs, resulting in lower VFAs concentrations. By the 10th day, the average VFAs concentration in $R(0.125)_{NBW}$ was 1033 mg/L, representing decrease of 20.8 % compared to $R(0.125)_{Control}$, indicating that NBW enhanced the consumption of VFAs after the hydrolysis stage.

The VFAs concentration in R(0.25) increased slowly, followed by a rapid decrease after 16th day. This suggests that the consumption of VFAs in R(0.25) was severely inhibited in the initial stage of AD. The rapid decrease thereafter suggests a gradual relief of VFAs inhibition in the later stages of AD. Throughout the AD process, $R(0.25)_{NBW}$ was significantly lower than $R(0.25)_{Control}$ on the 10th, 16th, and 22nd days, indicating that NBW accelerated VFAs consumption in the hydrolysis acidification stage. For $R(0.5)_{Control}/R(0.5)_{NBW}$, the VFAs concentration did not decrease during the AD process, indicating that NBW was unable to effectively alleviate severe VFAs inhibition.

3.2. Changes in stability parameters

The pH is an important index for measuring the stability. The pH values of the eight groups were measured periodically (Fig. 2a). For $R(0.5)_{Control}/R(0.5)_{NBW}$, the pH throughout the AD process was below 5.3, indicating severe VFAs inhibition. $R(0.25)_{Control}/R(0.25)_{NBW}$ experienced a slow increase followed by a rapid increase and then stabilised,

indicating that these two groups experienced severe VFAs inhibition, which subsequently eased. Additionally, the pH of $R(0.25)_{NBW}$ was higher than $R(0.25)_{Control}$ on the 4th, 10th, and 16th days, indicating that NBW may improve the stability, thus relieving VFAs inhibition. For the R(0.125), the pH gradually increased to approximately 7.5, indicating a minimal impact of VFAs inhibition. R(0) showed an initial decrease, followed by an increase and stabilisation in pH value. The initial decrease was due to the production of acidic substances during the acidification process.

Fig. 2b shows the TA content under different degrees of VFAs inhibition. The initial TA content of R(0) was the highest at approximately 1101 mg $CaCO_3$/L, followed by R(0.125) (approximately 895 mg $CaCO_3$/L), and R(0.25) and R(0.5) being the lowest at approximately 751 mg $CaCO_3$/L and 463 mg $CaCO_3$/L, respectively. $R(0)_{Control}$ and $R(0.125)_{Control}$ were less than their respective NBW groups on the 4th, 10th, and 16th days, implying that NBW was beneficial for increasing TA content when VFAs inhibition was absent or mild. Additionally, the TA content in R(0.5) was significantly lower than other six groups throughout the AD ($p < 0.05$), combined with a pH value below 5.3, indicating that severe VFAs inhibition occurred during the AD.

Fig. 2c depicts how PA content changes during the AD process. Similar to the trend in TA content, R(0), R(0.125), and R(0.25) groups showed an increasing trend in PA content during AD. On days 4 and 10, $R(0)_{Control}$ and $R(0.125)_{Control}$ were lower than $R(0)_{NBW}$ and $R(0.125)_{NBW}$, respectively. The difference on the 4th day was significant, implying that NBW promoted a rapid increase in PA content. Furthermore, the PA content of $R(0.25)_{NBW}$ was higher than $R(0.25)_{Control}$, indicating that NBW facilitated a rapid increase in PA content, thereby enhancing the buffering capacity and alleviating VFAs inhibition.

The results for the IA/PA are shown in Fig. 2d. Initially, the IA/PA value of R(0.25) was high, indicating poor stability owing to severe VFAs inhibition during the early AD process. In contrast, R(0) and R(0.125) were relatively small throughout the AD process, indicating a strong buffering capacity. It is worth mentioning that $R(0.25)_{NBW}$ was significantly higher than the respective control group on day 10, indicating that NBW could facilitate rapid decreases in IA/PA. The IA/PA values of R(0), R(0.125), and R(0.25) initially decreased, followed by stabilisation, indicating a strong buffering capacity in the later stages of AD.

Fig. 2e shows the influence of NBW on VFA/TA under different degrees of VFAs inhibition. To maintain a healthy CH_4 production in an anaerobic digester, the ratio of VFA/TA should be in the range of 0.1 to 0.35 (Yuan et al., 2009). For $R(0.5)_{Control}/R(0.5)_{NBW}$, the VFA/TA values were all greater than 6.9 throughout the AD process, indicating severe VFAs inhibition and an extremely weak buffering capacity. Although $R(0.25)_{Control}/R(0.25)_{NBW}$ exceeded 4.0 in the beginning, as the AD process progressed, its VFA/TA value gradually decreased and eventually became less than 0.1, indicating that although it experienced severe VFAs inhibition in the beginning, the inhibition gradually decreased in the later stages of AD. The trends of R(0) and R(0.125) were similar to that of R(0.25), showing gradual downward trends. The values of R(0.125) and R(0) were low, indicating mild or no significant VFAs inhibition. Additionally, $R(0)_{Control}$ was higher than $R(0)_{NBW}$ on day 4, and $R(0.125)_{Control}$ and $R(0.25)_{Control}$ were higher than their respective control groups on the 10th and 16th days, indicating that NBW facilitated rapid decreases in VFA/TA, shortened the lag phase, and enhanced the buffering capacity. The trends in VFA/TA were similar to those of IA/PA.

In summary, when severe VFAs inhibition occurred during the AD process, such as in $R(0.5)_{Control}/R(0.5)_{NBW}$, NBW did not significantly improve the AD process stability. When VFAs inhibition was not severe, NBW can improve system stability and buffer capacity. Further studies are needed to determine the threshold for VFAs inhibition under NBW conditions.

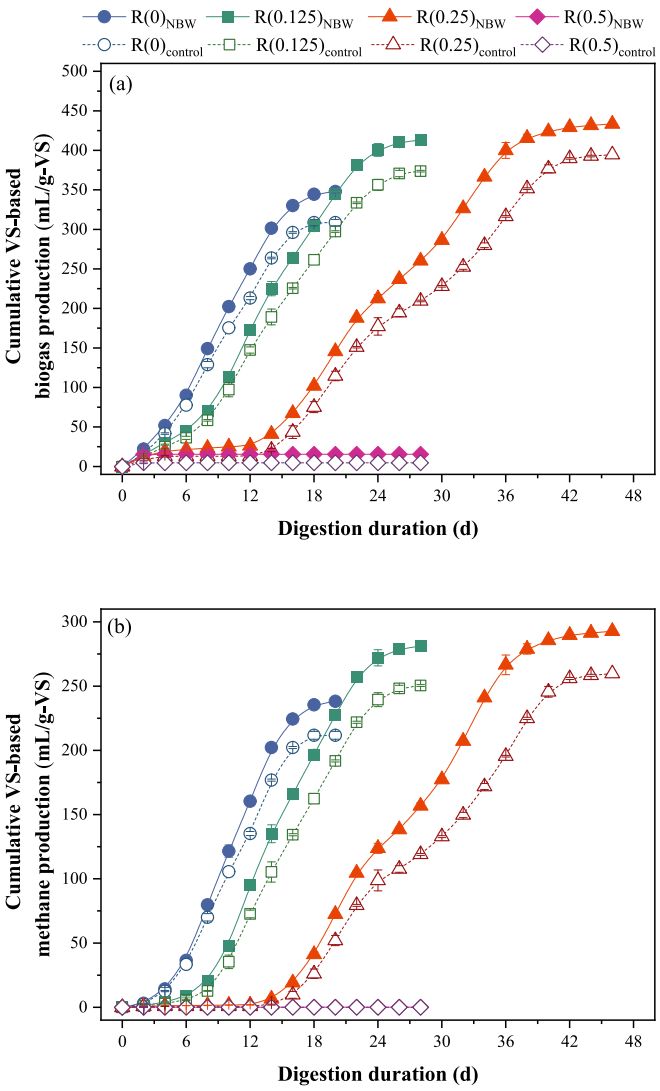


Fig. 3. Cumulative VS-based biogas and methane production. (a) Cumulative VS-based biogas production. (b) Cumulative VS-based methane production.

3.3. Performance of biogas and CH_4 production

As shown in Fig. 3, the average cumulative biogas production values of $R(0.25)_{NBW}$, $R(0.125)_{NBW}$, and $R(0)_{NBW}$ were 433, 413, and 348 mL/g-VS, respectively, representing increases of 10.0 %, 10.7 %, and 12.9 %, respectively, compared to their corresponding control groups. Moreover, the difference of biogas production between R_{NBW} and $R_{Control}$ was significant ($p < 0.05$). The AD processes were severely inhibited in $R(0.5)_{Control}/R(0.5)_{NBW}$, and NBW was unable to effectively alleviate VFAs inhibition.

For cumulative CH_4 production, $R(0.25)_{NBW}$ obtained the highest relative cumulative CH_4 production, averaging 293 mL/g-VS, representing increases of 13.1 % compared to $R(0.25)_{Control}$. The cumulative CH_4 production of $R(0.125)_{NBW}$ and $R(0)_{NBW}$ was 281 and 238 mL/g-VS, respectively, representing increases of 12.2 % and 12.3 %, respectively, compared to their corresponding control groups. In this study, CH_4 production and increment obtained from $R(0)_{NBW}$ (compared to $R(0)_{control}$) were consistent with results obtained from previous studies using Air-NBW in AD of swine manure (Fan et al., 2020). González-García et al., (2021) found that simultaneous ammonia recovery using gas-permeable membranes improves the AD of swine manure and increases CH_4 yield to an average of 10–17 %. Yang et al., (2021) found that under the optimal biochar dosage of 5–10 %, CH_4 yield was

Table 2
The economic assessment when treating 1 ton of swine manure in this study.

Reactors	CH_4 production (L)	Heating value of CH_4 (kJ) Electricity generation (kWh)
$R(0)_{Control}$	48,324	1,934,410 537.34
$R(0)_{NBW}$	54,346	2,175,470 604.30

The unit conversion from kJ to kWh is 1/3600.

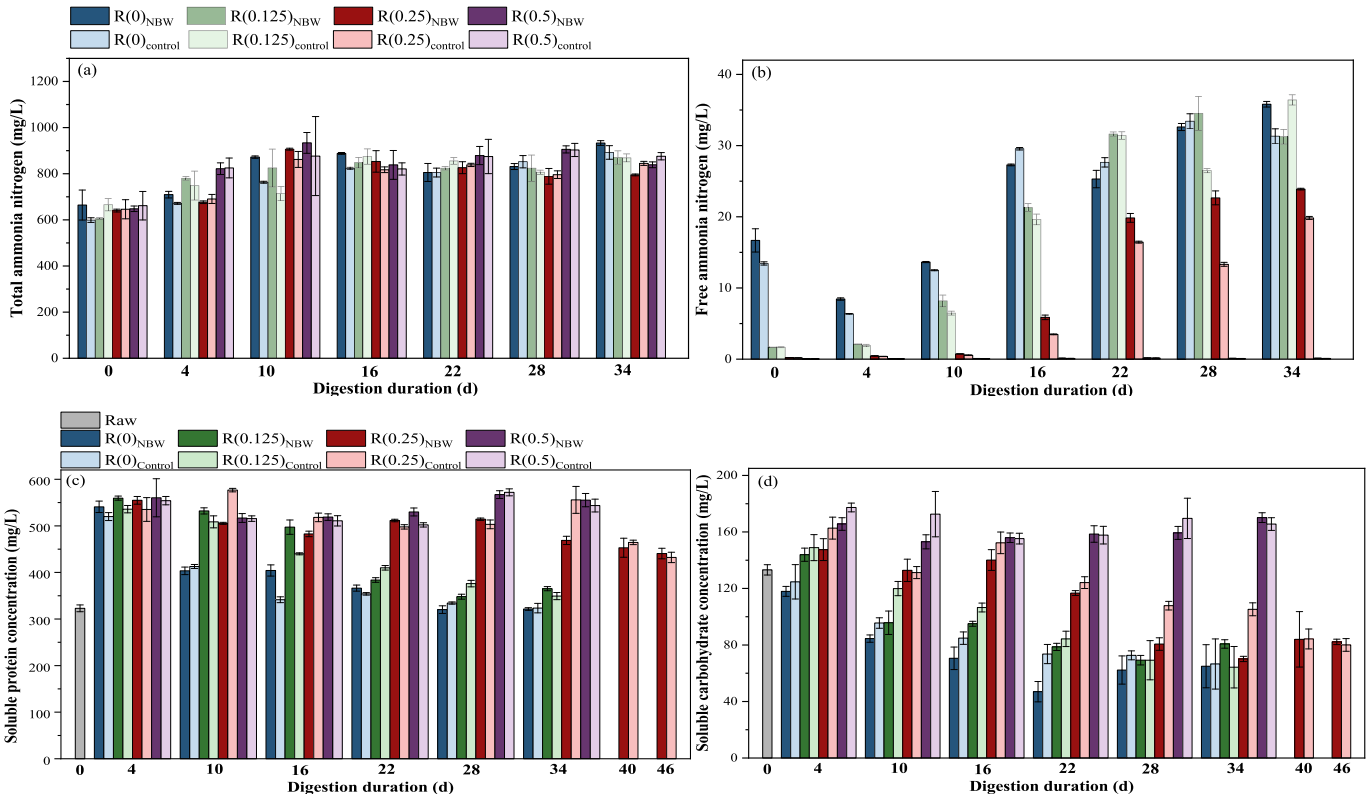


Fig. 4. Performance of TAN, FAN, and soluble organics. (a) Total ammonia nitrogen. (b) Free ammonia nitrogen. (c) Soluble protein concentration. (d) Soluble carbohydrate concentration.

significantly improved by 25 % in AD of swine manure. Therefore, the supplementation of NBW can also effectively increase CH_4 production during AD of swine manure compared to other methods. Moreover, this method does not require too much energy or complex devices, and does not consume chemicals or produce pollution. Due to severe VFAs inhibition, CH_4 production was severely affected in R(0.5). Moreover, the difference of CH_4 production between R_{NBW} and $R_{Control}$ was significant ($p < 0.05$). Economic assessment was used in this study, where the heating value of CH_4 is 40.03 kJ/L under 1 atm and 25 °C (Silva et al., 2018). The results showed that the supplementation of NBW resulted in 604.30kWh of electricity generated by processing one ton of swine manure in this study, which was significantly better than the control group (Table 2).

It is worth mentioning that R(0.125) were higher than those of R(0), which was likely due to the supplementation of VFAs in R(0.125) providing organic acids for the AD process and increasing the nutrient content available for microorganisms. Interestingly, although $R(0.25)_{NBW}$ exhibited the highest cumulative biogas and CH_4 production, its increase was less than of $R(0.125)_{NBW}$, which was insufficient to compensate for the doubling of the VFAs addition, indicating that severe VFAs inhibition was detrimental to biogas and CH_4 production.

3.4. Performance of TAN, FAN, and soluble organics

In this study, TAN content was measured to clarify whether ammonia inhibition occurs during the AD process and whether it synergistically interacts with VFAs inhibition. As shown in Fig. 4a, the TAN content in R_{NBW} and $R_{Control}$ were less than 1200 mg/L. Therefore, none of the groups in this study experience ammonia inhibition. The FAN content in R_{NBW} and $R_{Control}$ were less than 40 mg/L (Fig. 4b), indicating that none of the groups experienced FAN inhibition according to previous reports on the threshold for FAN (De Baere et al., 1984). Furthermore, because the pH values of $R(0.5)_{Control}$/$R(0.5)_{NBW}$ were less than 5.3, the FAN contents of these two groups were minimal. Overall, no inhibition of ammonia was observed in this study.

Fig. 4c shows the performance of the soluble proteins. Except for R (0.5), the soluble proteins of the other six groups initially increased, then decreased, and eventually stabilised. This increase was most likely caused by hydrolysis, which increased soluble protein production. Compared to R(0) and R(0.125), the decrease in soluble proteins in R (0.25) was slower, possibly because VFAs inhibition partially hindered the consumption of soluble proteins. The soluble proteins of $R_{Control}$ were lower than their respective R_{NBW} on the 4th day, indicating that NBW accelerated the production of soluble proteins. The trends in soluble carbohydrates were similar to those in soluble proteins (Fig. 4d). Due to severe VFAs inhibition, R(0.5) was not significantly consumed. The soluble carbohydrates of R(0.125) and R(0.25) increased initial, then decreased, and eventually stabilised. R(0) gradually declines and then stabilises.

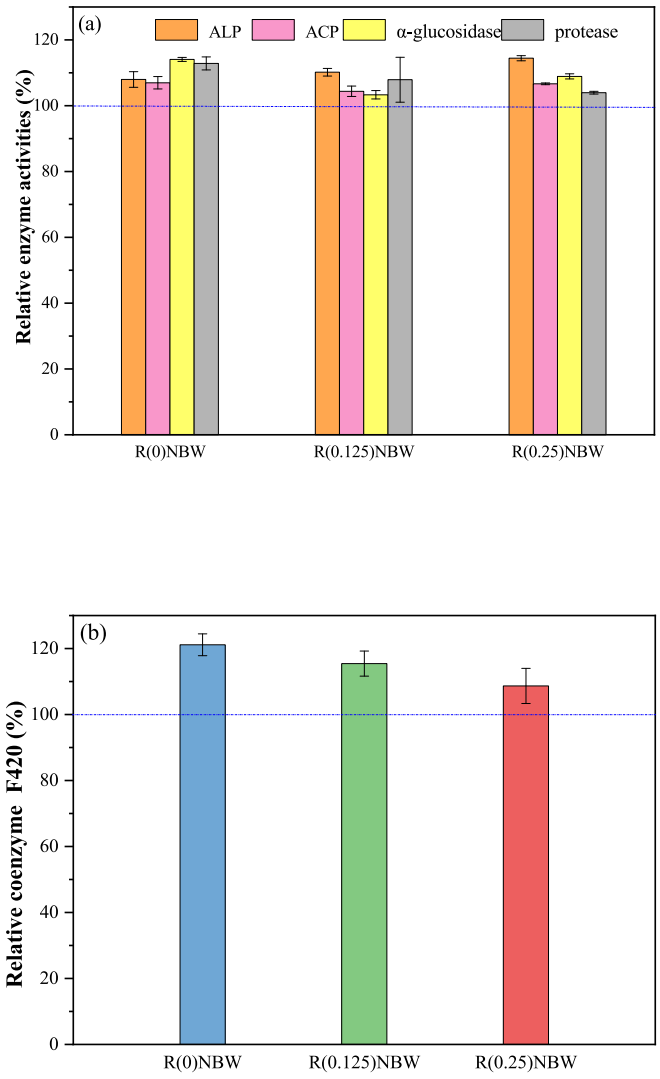


Fig. 5. Enzyme activity performance. (a) Relative activities of extracellular hydrolases. (b) Relative activities of Coenzyme F420.

3.5. Enzyme activity performance

Enzymes are essential components of the AD process, serving important functions throughout the process. This study investigated the activities of enzymes in two limiting steps of AD: hydrolysis and methanogenesis. A comparative study was conducted on extracellular hydrolytic enzymes such as ACP, ALP, α-glucosidase, and proteases. Fig. 5a shows that the four extracellular hydrolytic enzymes in R_{NBW} were more active than $R_{Control}$ with enzyme activity increased by 3.3 % to 15.2 % in different groups. This phenomenon indicated that NBW enhanced the activity of hydrolytic enzymes, thereby accelerating the hydrolysis process under VFAs inhibition. Subsequently, F420, which reflects the activity of methanogens, was evaluated in this work. The results showed that F420 in R_{NBW} were more active than $R_{Control}$ with enzyme activity increased by 8.7 % to 21.1 % in different groups (Fig. 5b). Therefore, NBW can enhance the activity of extracellular hydrolytic enzymes and coenzyme F420, thereby accelerating the hydrolysis and methanogenic stage.

3.6. Microbial community structure

To uncover the underlying mechanisms and gain a deeper understanding of the microbial community structure under different degrees of VFAs inhibition, we performed high-throughput sequencing analysis. The archaeal and bacterial diversity results are shown in Table 3 and 4. The coverage indices exceeded 0.99, indicating that the sequencing data covered almost all the sequences in the samples.

In archaeal sequencing, as indicated by the Chao and Ace indices, the abundances of archaeal microbial communities in R_{NBW} were higher than those in control groups, indicating that NBW was conducive to improving the performance of archaeal microbial communities in the AD process. Additionally, as VFAs inhibition intensified, the abundance of archaeal microbial communities in R_{NBW} gradually increased, and the increase was relatively larger than that in the corresponding control groups, implying that NBW helped alleviate VFAs inhibition on archaea. Furthermore, the Shannon and Simpson indices of $R(0.25)_{NBW}$ were the highest, indicating that the archaeal microbial diversity of $R(0.25)_{NBW}$ was the highest and further demonstrating the positive effects of NBW on archaeal biodiversity.

The dominant phyla were *Abditibacteriota*, *Acidobacteriota*, *Actinobacteriota*, and *Bacteroidetes* (Fig. 6a), which ferment polysaccharides in the AD systems and participate in the degradation of macromolecular organics (Strąpoć et al., 2011). At the genus level, *Sulfurovum* belongs to autotrophic facultative sulfur-oxidizing bacteria, which can oxidize sulfides using oxygen or nitrate as electron acceptors to produce energy (Yan et al., 2023b). Sulfur-oxidizing bacteria and methanogens complement each other in nutritional metabolism (Nie et al., 2021). *Sulfurovum* showed an initial decreasing trend, followed by an increase with increasing VFAs concentrations (Fig. 6b), indicating that high VFAs concentrations are unfavorable for CH_4 production. The genus *f_bacteroidetes_vadinHA17* is a facultative anaerobic denitrifying bacterium capable of degrading various complex organic compounds in AD systems. Studies have shown that an increase in *f_bacteroidetes_vadinHA17* content can ferment more complex organic compounds, such as steroids, to accelerate reaction rates (Baker et al., 2020). Fig. 6a shows that *f_bacteroidetes_vadinHA17* was higher in R_{NBW} than $R_{Control}$, which could explain why NBW increased hydrolysis rate and CH_4 production.

Hydrogenophilaceae are denitrifying bacteria that play important roles in nitrogen reduction. Compared with the control group, *Hydrogenophilaceae* in R_{NBW} was enriched to a certain extent. With an increase in VFAs concentration, *Hydrogenophilaceae* in R_{NBW} gradually decreased, indicating that NBW was conducive to *Hydrogenophilaceae* in the AD process when VFAs inhibition was not severe. Additionally, *Anaerolineaceae*, a representative family of Chloroflexi, not only has denitrification functions but also participates in the degradation of carbohydrates (Sun et al., 2015). *Anaerolineaceae* initially increased and then decreased with increasing VFAs concentrations in both $R_{Control}$ and R_{NBW}, indicating that severe VFAs inhibition was unfavourable for

Table 3
Analysis results of archaea diversity in this work.

Indices	$R(0)_{Control}$	$R(0)_{NBW}$	$R(0.125)_{Control}$	$R(0.125)_{NBW}$	$R(0.25)_{Control}$	$R(0.25)_{NBW}$
Feature	602	702	575	987	1063	1849
ACE	641.0288	770.3177	627.4916	1047.417	1114.063	1890.581
Chao1	617.682	731.2742	596.322	1008.302	1081.361	1857.863
Simpson	0.9549	0.9581	0.954	0.9557	0.9557	0.9704
Shannon	5.5012	5.6284	5.3839	5.6256	5.7012	6.6873
Coverage	0.9979	0.996	0.9968	0.9976	0.9976	0.9976

Table 4
Analysis results of bacterial diversity in this work.

Indices	$R(0)_{Control}$	$R(0)_{NBW}$	$R(0.125)_{Control}$	$R(0.125)_{NBW}$	$R(0.25)_{Control}$	$R(0.25)_{NBW}$
Feature	1886	2213	1984	1868	2221	1964
ACE	1905.032	2234.95	2010.895	1888.891	2239.846	1985.964
Chao1	1888.849	2215.365	1988.974	1870.138	2222.404	1966.474
Simpson	0.9027	0.9735	0.9515	0.932	0.9784	0.9527
Shannon	7.2015	8.3025	7.7706	7.4005	8.459	7.8676
Coverage	0.9992	0.9991	0.9989	0.999	0.9991	0.9991

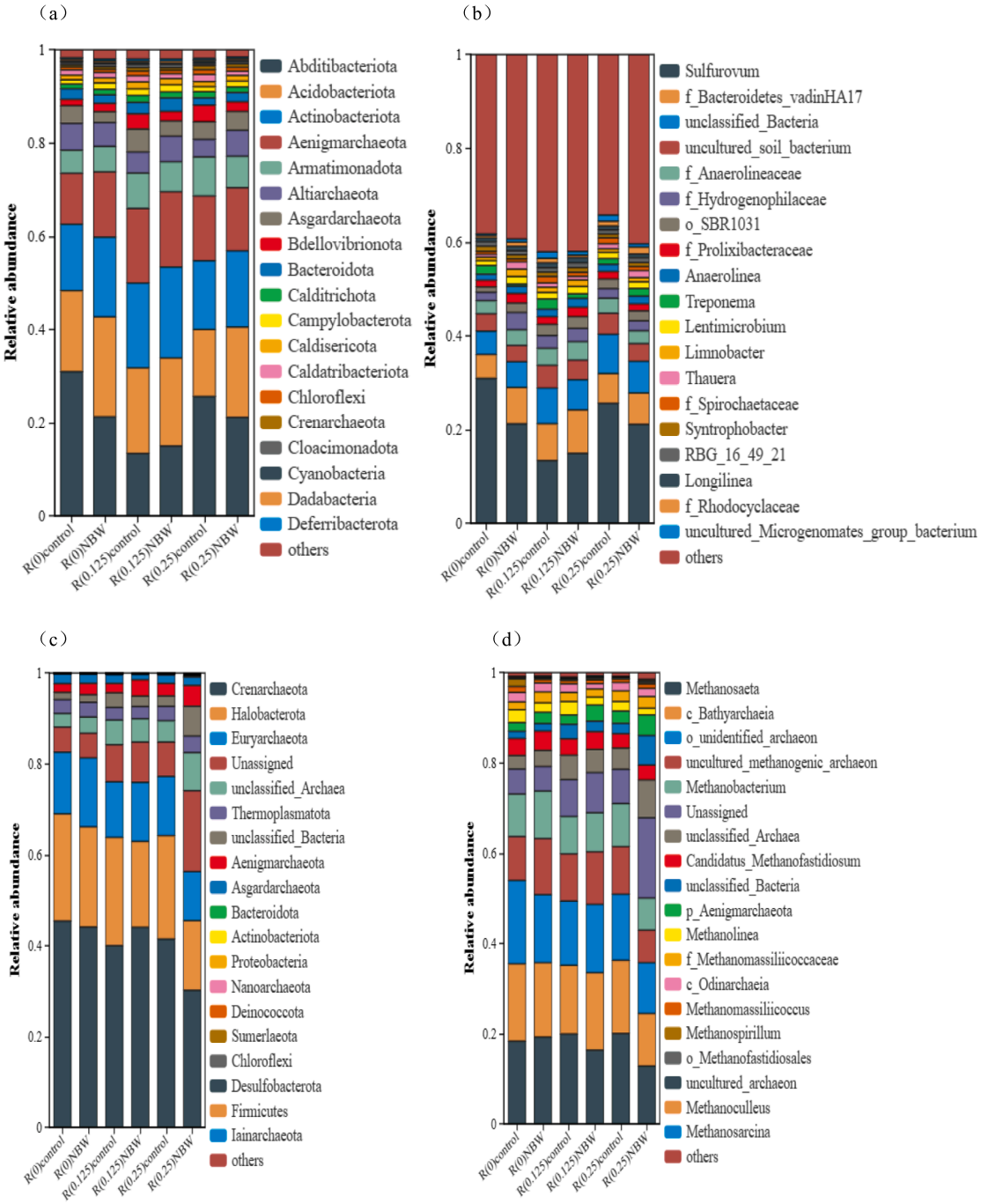


Fig. 6. Bacterial and archaea community analysis. (a) Phylum level of bacteria. (b) Generic level of bacteria. (c) Phylum level of archaea (d) Generic level of archaea.

denitrification. Furthermore, the proportion of *Anaerolineaceae* was highest in $R(0.125)_{NBW}$, indicating that NBW could promote denitrification.

In conclusion, the bacterial community analysis revealed that *Sulfurovum* and *f_bacteroidetes_vadinHA17* were the dominant bacterial genera, contributing significantly to the conversion of sulfides and the

degradation of complex organics. *Hydrogenophilaceae* and *Anaerolineaceae* were the primary contributors to denitrification, and NBW may have aided CH_4 production and denitrification. Although high concentrations of VFAs have inhibitory effects, NBW alleviated VFAs inhibition to some extent and promoted denitrification.

Crenarchaeota accounted for the largest proportion, with relative abundances ranging from 30.10 % to 40.50 %. *Halobacterota* was the second most dominant phylum, accounting for approximately 16.80 % to 23.50 %, whereas *Euryarchaeota* accounted for approximately 9.50 % to 14.90 %. The dominant archaeal genera were *Methanosaeta*, *c_Bathyarchaeia*, *o_unidentified_archaeon*, *uncultured_methanogenic_archaeon, Methanobacterium*, and *Candidatus_Methanfastidiosum* (Fig. 6d). *Methanosaeta* is a common acetate-type CH_4-producing bacterium in CH_4-producing systems, and it was enriched in all groups. Additionally, *Methanobacterium* is a hydrogenotrophic methanogen (HM), whose primary sources of energy are CO_2 and H_2. It was more enriched in R_{NBW} than in their respective control groups, indicating that both NBW and control groups relied primarily on the acetate-type CH_4 production pathway and that NBW improved the HM metabolic pathway.

To conclude, the three dominant phyla involved in CH_4 production *Halobacterota, Euryarchaeota, and Crenarchaeota*. *Methanosaeta* was relatively abundant across the genus level, indicating that acetate-type CH_4 production was the primary pathway. Furthermore, *Methanobacterium* was more enriched in the NBW groups, suggesting that NBW improved the HM metabolic pathway.

4. Conclusion

This study demonstrated that the inhibition of VFAs on manure AD could be mitigated through the supplementation of NBW, which enhanced buffering capacity and CH_4 production. Mechanism analysis indicated that NBW improved the consumption of VFAs, thereby alleviating the inhibitory effects caused by VFA accumulation. Additionally, NBW facilitated a rapid increase in pH, TA and PA during the early stages of the AD process, along with a swift decrease in the IA/PA and VFA/TA ratios. Moreover, NBW significantly increased CH_4 production under VFAs inhibition. For microorganisms and enzymes, NBW elevated the activity of key enzymes (including 4 hydrolytic enzymes and coenzyme F420) under VFAs inhibition, enriched microbial community abundance, and strengthened HM metabolic pathways. These findings indicate the potential of NBW technology to serve as an alternative method for alleviating inhibition in agricultural AD systems. This study innovatively proposes the application of NBW to alleviate VFAs inhibition, which is common in AD of agricultural waste. This study highlights the potential of NBW to reduce VFAs inhibition and improve the efficiency of agricultural waste treatment. Given these promising results, NBW technology may hold significant potential for future applications in AD processes under VFAs inhibition and warrants further research.

CRediT authorship contribution statement

Yujie Fan: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ziyang Zhang:** Resources, Investigation, Data curation. **Xiaojing Yang:** Writing – review & editing, Conceptualization. **Haibo Yang:** Validation, Formal analysis. **Peng Deng:** Validation, Formal analysis. **Ziwen Zhao:** Writing – review & editing, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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